

Phase 1: The Dynamical System

First, we define the system of ordinary differential equations (the Lorenz system) and its solution trajectory ϕ . Assume the standard Lorenz parameters $r, \sigma, b > 0$.

$$\begin{aligned}\dot{x} &= \sigma(y - x) \\ \dot{y} &= rx - y - xz \\ \dot{z} &= xy - bz\end{aligned}$$

Let $\phi : \mathbb{R} \rightarrow \mathbb{R}^3$ be a function such that ϕ satisfies the ODE system above. Because ϕ satisfies these differential equations, it is continuously differentiable (C^1) by definition.

Phase 2: Defining V and $\tilde{V}$

Next, we define the functions V and $\tilde{V}$ and establish their relationship along the trajectory $\phi(t)$.

1. Define $V(x, y, z)$:

$$V(x, y, z) = rx^2 + \sigma y^2 + \sigma(z - 2r)^2$$

- **Prove:** V is continuous and differentiable (as it is a polynomial).

2. Define $\tilde{V}(x, y, z)$:

$$\tilde{V}(x, y, z) = -2\sigma(rx^2 + y^2 + bz^2 - 2brz)$$

- **Prove i:** $\tilde{V}$ is continuous (as it is a polynomial).
- **Prove ii:** Because both V and ϕ are differentiable (C^1), we can apply the chain rule to show that the time derivative of V along the flow ϕ is exactly $\tilde{V}$.

$$\frac{d}{dt}(V \circ \phi) = \tilde{V} \circ \phi$$

Phase 3: The Set D and Supremum c

We now define a region D based on $\tilde{V}$ and show that V reaches a maximum value on it.

3. Define the set D :

$$D := \{\tilde{V} \geq 0\}$$

- **Prove i:** D is compact.
 - **Closed:** D is closed via the continuity of $\tilde{V}$.
 - **Bounded:** D is bounded because there exists some radius $r > 0$ such that for all $|(x, y, z)| > r$, we have $\tilde{V}(x, y, z) < 0$. Since D is closed and bounded in $\mathbb{R}^3$, it is compact.
- **Prove ii:** V attains a supremum c on D .

$$\sup_D V = c < \infty$$

This is guaranteed because V is a continuous function evaluated on a compact set D .

Phase 4: The Set E and its Boundary Properties

We define a set E slightly larger than the maximum value c , and analyze its boundary ∂E .

4. Define some $\epsilon > 0$ and the set E :

$$E := \{V \leq c + \epsilon\}$$

- **Prove i:** E is compact.
 - **Closed:** E is closed via the continuity of V .
 - **Bounded:** E is bounded because there exists some radius $r > 0$ such that for all $|(x, y, z)| > r$, we have $V(x, y, z) > c + \epsilon$. Since E is closed and bounded in $\mathbb{R}^3$, it is compact.
- **Prove ii:** The boundary of E is contained in the level set $\{V = c + \epsilon\}$.

$$\partial E \subseteq \{V = c + \epsilon\}$$

- **Argument:** Let $w \in \partial E$. Since E is closed, $\partial E \subset E$, which implies $V(w) \leq c + \epsilon$.
- Assume for contradiction that $V(w) < c + \epsilon$.
- There exists an open interval $U = (a, b)$ such that $a < V(w) < b < c + \epsilon$.
- Because V is continuous, the preimage $V^{-1}(U)$ is open.
- We know $V^{-1}(U) \subset E$, which implies $V^{-1}(U) \subset \text{int}(E)$.
- Since $w \in V^{-1}(U)$, this means $w \in \text{int}(E)$.

- This is a contradiction ($\perp$), because a point cannot be in both the boundary ∂E and the interior $\text{int}(E)$. Therefore, it must be that $V(w) = c + \epsilon$.

- **Prove iii:** The boundary of E is disjoint from D .

$$\partial E \cap D = \emptyset$$

- **Argument:** For any $w \in \partial E$, we just proved $V(w) = c + \epsilon$. Since $\epsilon > 0$, we have $c + \epsilon > c$. But $c = \sup_D V$. If w were in D , $V(w)$ could not be strictly greater than the supremum c . Therefore, $w \notin D$.

- **Prove iv:** $\tilde{V}$ is strictly negative on the boundary of E .

$$\tilde{V} < 0 \text{ on } \partial E$$

- **Argument:** From step (iii), $\partial E \cap D = \emptyset$, meaning ∂E is entirely contained in the complement of D (D^c). Since $D = \{\tilde{V} \geq 0\}$, its complement D^c is exactly the set of points where $\tilde{V} < 0$. Since $\partial E \subset D^c$, it follows that $\tilde{V} < 0$ on ∂E .

Phase 5: Trapping Region and Positive Invariance

Theorem: If $\phi(t_0) \in E$, then $\phi(t) \in \text{int}(E)$ for all $t > t_0$.

Case 1: $\phi(t_0) \in \text{int}(E)$ Assume on the contrary that the trajectory does not remain in the interior. Then there exists some $t > t_0$ such that $\phi(t) \notin \text{int}(E)$.

Let $S = \{t > t_0 \mid \phi(t) \notin \text{int}(E)\}$. We define τ to be the infimum of this set:

$$\tau := \inf S$$

This infimum exists because S is bounded below (by t_0) and is non-empty by our contradictory assumption. We now evaluate the position of $\phi(\tau)$, which must fall into one of three cases.

Subcase 1: $\phi(\tau) \in \text{int}(E)$

- By the definition of an infimum, there exists a sequence $t_n \rightarrow \tau$ such that $t_n \in S$.
- Therefore, $\phi(t_n) \notin \text{int}(E)$, which means $\phi(t_n) \in (\text{int}(E))^c$ for all n .
- Because ϕ is continuous, the limit of the sequence is $\phi(\tau) = \lim_{n \rightarrow \infty} \phi(t_n)$.
- Since the complement $(\text{int}(E))^c$ is a closed set, it contains all its limit points. Thus, the limit $\phi(\tau)$ must also be in $(\text{int}(E))^c$.
- This contradicts our assumption for this subcase that $\phi(\tau) \in \text{int}(E)$. ($\perp$)

Subcase 2: $\phi(\tau) \in E^c$

- **By a lemma proven previously, since ϕ is a continuous function, the interval $[t_0, \tau]$ is connected, and E is connected, any path from $\text{int}(E)$ to E^c must intersect the boundary ∂E .**
- Thus, there must exist some time t'' such that $t_0 < t'' < \tau$ where $\phi(t'') \in \partial E$.
- Because ∂E is disjoint from $\text{int}(E)$, $\phi(t'') \notin \text{int}(E)$, meaning $t'' \in S$.
- But $t'' < \tau$, which contradicts the definition of τ as the infimum of S . ($\perp$)

Subcase 3: $\phi(\tau) \in \partial E$

- By the definition of τ , for all t strictly between t_0 and τ ($t < \tau$), the trajectory is still in the interior: $\phi(t) \in \text{int}(E)$.
- Therefore, $V(\phi(t)) \leq c + \epsilon$ for all $t < \tau$.
- Since $\phi(\tau) \in \partial E$, we know from earlier that $V(\phi(\tau)) = c + \epsilon$.
- **Because ϕ and V were established earlier as continuously differentiable (C^1), the true two-sided derivative exists. This guarantees that the left-hand limit is exactly equal to the derivative at τ :**

$$\left. \frac{d}{dt} V(\phi(t)) \right|_{t=\tau} = \lim_{t \rightarrow \tau^-} \frac{V(\phi(\tau)) - V(\phi(t))}{\tau - t}$$

- Because $V(\phi(\tau)) = c + \epsilon$ and $V(\phi(t)) \leq c + \epsilon$, the numerator is ≥ 0 . Because $t \rightarrow \tau$ from below, $\tau - t > 0$, so the denominator is positive.
- Thus, the limit evaluates to a non-negative value:

$$\left. \frac{d}{dt} V(\phi(t)) \right|_{t=\tau} \geq 0$$

- However, we established earlier that $\frac{d}{dt}(V(\phi(t))) = \tilde{V}(\phi(t))$, and $\tilde{V} < 0$ on ∂E .
- This gives $0 \leq \tilde{V}(\phi(\tau)) < 0$, which is a contradiction. ($\perp$)

Case 2: $\phi(t_0) \in \partial E$. Assume the contrary: there exists some time $t' > t_0$ such that $\phi(t') \notin \text{int}(E)$. This leaves us with two possible subcases for the trajectory's behavior between t_0 and t' .

Subcase 1: There is an intermediate point in the interior.

- Assume there exists a time t'' strictly between t_0 and t' ($t_0 < t'' < t'$) such that $\phi(t'') \in \text{int}(E)$.
- By the proof from Case 1, once the trajectory is in the interior at time t'' , it must remain in the interior for all subsequent times $t > t''$.
- Since $t' > t''$, this strictly implies that $\phi(t') \in \text{int}(E)$.
- This directly contradicts our initial assumption that $\phi(t') \notin \text{int}(E)$. ($\perp$)

Subcase 2: There is no intermediate point in the interior.

- Assume there is no such $t'' \in (t_0, t']$ where $\phi(t'') \in \text{int}(E)$.
- This implies that for all t in the interval $(t_0, t']$, the trajectory never enters the interior: $\phi(t) \notin \text{int}(E)$.
- Consequently, for all $t \in (t_0, t']$, the trajectory must be either on the boundary ∂E or in the complement E^c .
- In both of these regions, the scalar function V satisfies $V(\phi(t)) \geq c + \epsilon$.
- We are given our initial condition for this case: $\phi(t_0) \in \partial E$. This means exactly $V(\phi(t_0)) = c + \epsilon$.
- We evaluate the right-hand limit definition of the time derivative of $V \circ \phi$ at $t = t_0$:

$$\lim_{t \rightarrow t_0^+} \frac{V(\phi(t)) - V(\phi(t_0))}{t - t_0}$$

- For $t \in (t_0, t']$, we have $V(\phi(t)) \geq c + \epsilon$, and $V(\phi(t_0)) = c + \epsilon$. Therefore, the numerator is ≥ 0 .
- Because we are approaching t_0 from the positive side ($t \rightarrow t_0^+$), the denominator $t - t_0$ is strictly positive.
- Thus, the right-hand limit must evaluate to a non-negative value (≥ 0).
- **Because ϕ and V were established earlier as continuously differentiable (C^1), the true two-sided derivative exists. This guarantees that the right-hand limit is exactly equal to the derivative at t_0 :**

$$\left. \frac{d}{dt} V(\phi(t)) \right|_{t=t_0} = \lim_{t \rightarrow t_0^+} \frac{V(\phi(t)) - V(\phi(t_0))}{t - t_0} \geq 0$$

- However, we established earlier that $\frac{d}{dt}(V \circ \phi) = \tilde{V} \circ \phi$. Since $\phi(t_0) \in \partial E$, we proved that $\tilde{V}(\phi(t_0)) < 0$.
- This yields $0 \leq \left. \frac{d}{dt} V(\phi(t)) \right|_{t=t_0} = \tilde{V}(\phi(t_0)) < 0$, which is a contradiction. ($\perp$)